GOBI Fieldbook

APPN Aerial SOP — Locked revision 1.0

APPN – GOBI M350 Fieldbook

This fieldbook provides a standardised operational guide for APPN GOBI UAV deployments on the DJI M350, supporting safe flight operations, consistent sensor configuration, and high-integrity data capture. It is intended for trained APPN staff conducting hyperspectral, LiDAR, RGB, and GNSS-INS data acquisitions, and promotes repeatability, transparency, and confidence in downstream data analysis across APPN operations.

Important

This protocol must be followed for all standard APPN GOBI UAV flights. Adherence to these procedures is essential to ensure operational safety, data integrity, and comparability of datasets across deployments. For any flights that **fall outside standard operating procedures**, detailed records must be kept documenting all deviations, including the specific settings changed, the rationale for those changes, and any anticipated implications for data quality or analysis.

Equipment Checklists

Note

Ensure batteries for all equipment are fully charged before heading to the field. Ensure charging cables are available for necessary equipment.

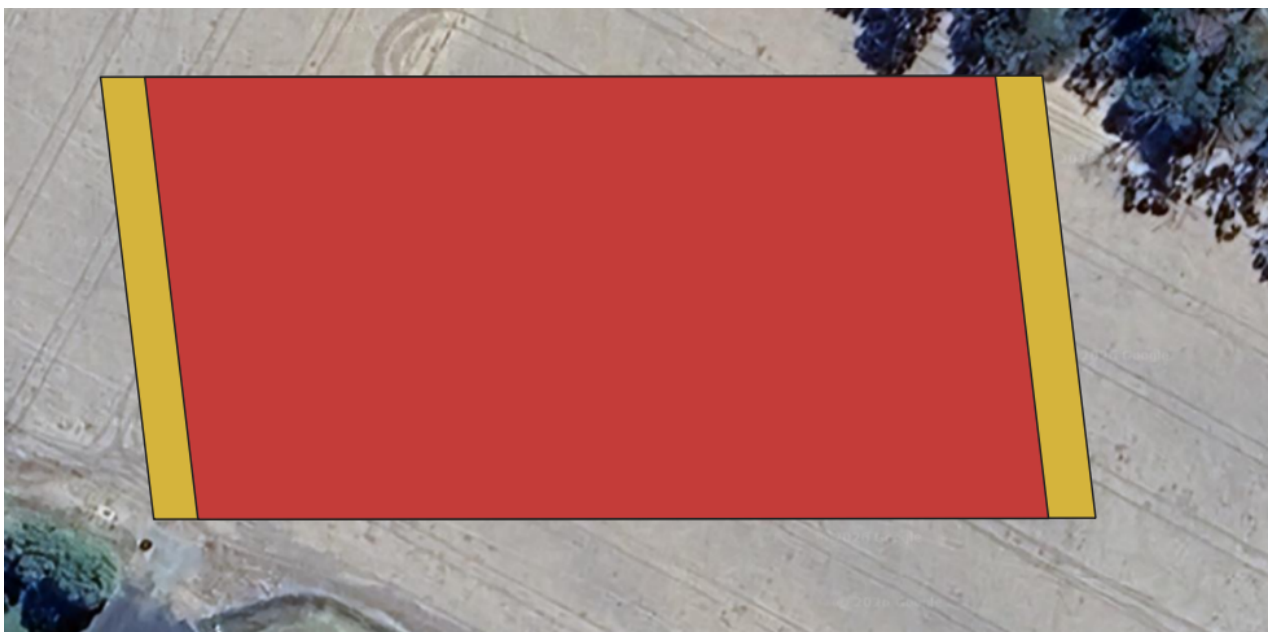
- ☐ **Aircraft**
  - ☐ Aircraft batteries
  - ☐ Landing gear
  - ☐ Landing pad
  - ☐ Spare parts
  - ☐ Tools
  - ☐ Logbook
- ☐ **Radio Control Transmitter / Ground Control Station**
- ☐ **GRYFN Gobi**
  - ☐ Gobi system
  - ☐ Power cables, chargers
  - ☐ Ethernet cable
  - ☐ SD card
  - ☐ *Capture* polygon files
- ☐ **Ground reference kit**
  - ☐ Reflectance calibration panels (11%, 30%, 56%, 82%)
  - ☐ Calibration validation panels (20%, 45%)
  - ☐ Ground control points and RTK GNSS system
  - ☐ 2 × folding tables to elevate panels
  - ☐ *If over 50 km from CORS base station*, a portable RTK base station (link to GRYFN gitbook)
- ☐ **Accessories**
  - ☐ Safety gear (signage and high-vis vests)

- ☐ Aeronautical radio
- ☐ Field laptop and spare batteries
- ☐ External storage media
- ☐ Water, food, esky, sunscreen, bug spray, first aid kit, etc.
- ☐ Spirit bubble, spirit level (or angle measurement) and measuring tape
- ☐ Portable fan (for cooling GOBI during data offload)
- ☐ External power brick (for charging UAV RC)

Flight Planning – DJI M350

> **Warning**
>
> Ensure that you apply for UAV flight approvals for locations and dates of flights well in advance.

1. Using a GPS survey system (Emlid, Trimble…) or GIS software, create a polygon of the area of interest with a 5-metre margin around the area of interest.
2. Using the GRYFN flight planning spreadsheet and DJI Pilot 2, set the parameters such as speed, overlap, and altitude. Orientation of the survey will be North–South in a paddock, or row-aligned for a plot trial.
3. Create two polygons: a *survey* polygon and a *capture* polygon. Ensure the *survey* polygon is 2× the flight speed larger than the *capture* polygon in the direction of travel (as per the figure below). Save both polygons uniquely.



4. If using QGIS, import the *capture* KML into Google Earth and immediately export it again. This is due to the required formatting Google Earth supplies.
5. Import the *capture* KML into the HPI Polygon Tool and export. This polygon sets the activation of the hyperspectral sensor within Hyperspec3.
6. Import the *survey* polygon into DJI Pilot 2 (M350), or into QGroundControl (IF1200).
7. Using both the flight planning app (DJI Pilot 2 or QGroundControl) and the GRYFN flight calculator, determine the speed, altitude, and frame period required to survey the area of interest.
8. Ensure the frame period is $> 20\%$ oversampled and the side overlap is $> 30\%$.

Flight Planning – Inspired Flight IF1200A

1. Using a GPS survey system (Emlid, Trimble…) or a GIS software, create a polygon of the area of interest. Make sure your polygon includes the areas where you will place your calibration panels and GCPs, with an additional 5 m buffer to avoid incomplete data.
2. Save this polygon twice as a KML — once as a *survey* polygon and once as a *capture* polygon.
3. If using QGIS, export the polygon as a KML (in Geometry, select *include z-dimension*, and ensure the CRS is set to WGS 84). Import the *capture* KML into the HPI Polygon Tool and export. This polygon sets the activation of the hyperspectral sensor within Hyperspec3.
4. Import the *survey* polygon into QGroundControl.
5. Using both QGroundControl and the GRYFN flight calculator, determine the speed, altitude, and frame period required to survey the area of interest.
   - Do not go below 2 m/s or above 8 m/s for stability reasons (GRYFN). Speeds greater than 8 m/s lead to excessive aircraft pitch and speeds less than 2 m/s accentuate the impacts of wind on aircraft stability and cause a visibly less smooth trajectory.
   - Altitude and speed will be tested and recommended from APEx results.
   - Ensure the frame period is at a minimum of 20% oversampling, the side overlap is $> 40\%$ for the SWIR sensor, and the *turnaround distance* is $2\times$ flight speed ($> 3\times$ at > 6 m/s).
6. Ensure flight lines are in the direction of planting (GRYFN).

Pre-Flight

1. Conduct airspace and weather checks. If collecting hyperspectral data, collection time must be within ± 2 hours of solar noon (refer: https://gml.noaa.gov/grad/solcalc/).
2. *(Optional)* Set up the Emlid RTK (install a peg or permanent GCP below the base station for recurring flights), let it run for at least 15 minutes, and start recording the RINEX file before flying.
3. Set up reflectance targets on the tables for calibration and validation. 2 sets of GRYFN reflectance panels should be used if available, at alternate ends of the capture, with the validation panel located centrally. Make sure you have GCPs installed (and coordinates recorded using Emlid) in the field for geometric calibration. (TO DO: ADD STANDARD FLIGHT LAYOUT md)
4. Set up a safe UAV RTH location, RTH altitude, and other geo-fencing settings on the respective drone.
5. Attach payload (Gobi or CALViS) to the aircraft:
   - Connect the power cable from aircraft to payload (non-standard payload bus only).
   - Connect both GNSS antenna cables to the correct ports (match A1 and A2).
   - Remove RGB and Nano HP lens caps.
   - Insert RGB SD card.
   - For M350, connect the C-type power cable to power the gimbal (make sure the B side of the cable is toward the outer side of the drone and the upper side of the light sensor).
   - For M350, activate the battery-powered fan (if warm ambient conditions). Avoid leaving GOBI powered on in stagnant air or high heat ($>$ ~35–38 °C).
6. Power on the radio controller; check battery status.
7. Launch DJI Pilot 2 on the M350, or QGroundControl on the IF1200 controller.
8. Review the flight plan, checking operational height.
9. Power on the aircraft; confirm connection to RC/GCS and battery status; ensure Remote ID is enabled (if flying the IF1200).
10. If you are using RTK (M350), check that you are receiving GNSS corrections. There should be more than 8 satellites for good correction (check the RTK fix in the M350 controller settings). The IF1200 does not use RTK, but ensure a minimum of 8 satellites and GPS.
11. Upload the flight plan to the aircraft (IF1200).

Sensor Configuration (TO DO: ADD DETAILS FROM CALVIS)

1. Place the exposure reference panel under the Nano HP lens; avoid casting shadows.

- Place the drone legs upon exposure-angle cones, ensuring a fixed angle is achieved each exposure setting. **(angle provided through SIF excellence)**

2. Connect to GOBI Wi-Fi or connect the ethernet cable to the payload. Set a static IP address for the sensor at `10.0.65.2` (or anything other than `50`, `100`, and `128`, and must be less than `255`).
3. Navigate to the GOBI WebUI at `10.0.65.50`.
4. Ensure a valid elevation is achieved.
5. Name the mission, using the convention `YYYYMMDD_XXXX` (`YYYY` = year; `MM` = month; `DD` = day, must be 2 digits; `XXXX` = a short reference or abbreviation for the job).
6. Click **VNIR Setup**.
7. Open the Flight Calculator and update altitude/speed; adjust the oversampling buffer (must be at least 20%, recommended setting is 30%). **(value may change through hyperspectral excellence)**
8. Adjust exposure until ~95% **(value may change through hyperspectral excellence)** spectral saturation at the **peak of the 90th percentile line** without exceeding the frame period.
   - The maximum distance from the VNIR HS sensor to the GOBI reflectance panel is 40 cm. This is calculated based on the FOV and a 20 cm panel size.
9. Use the lowest gain mode that still provides sufficient saturation.
10. Import the polygon KML or text file.
11. Place the lens cap on the Nano HP.
12. Click **Collect Dark Reference**; remove the lens cap afterwards.
13. Remove the exposure reference panel.
14. Press **Start Mission**.
15. Remove the ethernet cable and begin flight operations.

Flight Operations

1. Ensure the aircraft (M350) is in **Position** flight mode.
2. Sync the flight plan to the radio controller if applicable; double-check the flight plan.
3. Clear people/objects away from the UAV.
4. Notify crew/observers that takeoff is beginning.
5. Begin manual takeoff, check stick controls work, and then fly to ~12 m AGL; **do not exceed** the trigger altitude (20 m default) before mission start.
6. Perform dynamic alignment (one to two figure-8 patterns).
7. Enable autonomous mission.
8. Monitor UAV battery voltage/percentage in flight.
9. After mission, switch back to **Position** mode to regain manual control.
10. Lower to capture altitude and perform post-mission dynamic alignment (figure-8).
11. Land and disarm UAV.
12. Leave the UAV, transmitter, and sensor powered on; begin post-flight checks. Ensure data transfer is finished before turning off the sensor.
13. **Do not hot-swap the batteries.** Treat each flight as a new flight.

Post-Flight Sensor Configuration

1. Reconnect to GOBI Wi-Fi or ethernet cable.
2. Open a browser to `10.0.65.50` or refresh the UI.
3. Press **Stop Mission**.
4. Replace the Nano HP and RGB lens caps.
5. Open WinSCP or equivalent and confirm the data.

Post-Flight

1. Disconnect the ethernet cable from the payload.

2. Disconnect the aircraft batteries.
3. Disconnect the payload power cable.
4. Disconnect the payload GNSS cables.
5. Undo the mounting clamp, remove the payload, and place it in its case.

Data Confirmation

Sensor type	File count	File size	Notes
GNSS-INS	1 file	~1.5 MB per minute	A new file will be created if time rolls ov
LiDAR	1 per minute	~175 MB per file	First & last file may be smaller. Files ma
VNIR	Mission time (s) / Frame Period value ~ number of data cubes	Several GB even for short flights.	Check that `frameindex`, `imu_gps`, and `sett`
RGB	One image every 2 s* (by default)	~30–70 MB per image	Check the event file vs the number of im

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Offload Data

1. Open WinSCP or similar FTP software.
2. Connect to the GRYFN Gobi via `gryfn@10.0.65.50` (password: `gryfn`).
3. Raw GNSS & LiDAR PCAP location: `/data/{mission name}`.
4. VNIR data location: `/data/capturedData/captured/{dateTime}`.
5. GOBI logs location: `/data/gryfn.log.{date}`.
6. RGB images are stored on the camera SD card.
7. Download GNSS, LiDAR, logs, and RGB after each mission; download hyperspectral data later due to its size.
8. Clear data directories after download to avoid filling the 500 GB SSD.

Data storage, processing & validation

All paths below follow the APPN folder structure. Formal paths use the wiki's placeholder syntax; an example follows each.

1. Downloaded data should be stored in the correct `T0_raw` folder.

   Formal path:

   ```
   ./{Node}/
     {YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
     {YYYYSiteName[_F|C]}/
     {SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/
   ```

   Example:

   ```
   ./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/
   ```

2. Data from the GCP points should be saved in the `T0_raw/Vault` folder (location may change).

   Formal path:

   ```
   ./{Node}/
     {YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
     {YYYYSiteName[_F|C]}/
     {SensorPlatform}/{YYYYMMDD}/run_XX/T0_raw/Vault/
   ```

   Example:

   ```
   ./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T0_raw/Vault/
   ```

3. Important information and issues should be recorded in the `FieldNotes.txt` file.

   Formal path:

```
./{Node}/
  {YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
  {YYYYSiteName[_F|C]}/
  {SensorPlatform}/{YYYYMMDD}/FieldNotes.txt
```

Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/FieldNotes.txt
```

4. Per-run information (e.g. APEx test conditions) should be stored in `RunOverview.csv`.

   Formal path:

```
./{Node}/
  {YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
  {YYYYSiteName[_F|C]}/
  {SensorPlatform}/{YYYYMMDD}/RunOverview.csv
```

   Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/RunOverview.csv
```

> **Important**
>
> When data from a failed run is being kept (e.g. debugging with GRYFN), the `RunFailed` boolean column of `RunOverview.csv` must be set to `True`.

5. The bundled `.graw` should be saved in the same `T0_raw` folder.
6. The `.gpro` should be generated using the APPN GPT Pipeline (`.json` and wiki links TBD) and stored in the adjacent `T1_proc` folder.

   Formal path:

```
./{Node}/
  {YYYY_ProjectDesc[_I|E][_Researcher][_org]}/
  {YYYYSiteName[_F|C]}/
  {SensorPlatform}/{YYYYMMDD}/run_XX/T1_proc/
```

   Example:

```
./USYD_Narrabri/2025_SIFCal/2025IAWatson_F/GOBI/20250825/run_00/T1_proc/
```

7. Standard QA process should be performed following this guide.

Appendix

FTP & Service Reference

Service	Protocol	IP	Port	User	Password
GOBI	FTP	`10.0.65.50`	22	`gryfn`	`gryfn`
SBG	FTP	`10.0.65.100`	21	`operator`	*(none)*
Headwall polygon	HTTP	`apps.headwallphotonics.com`	—	—	—

Headwall Polygon Tool — Workflow

1. Import KML.
   - Must be in Google Earth format.
2. Export KML.
3. Save KML.
4. In File Explorer, rename the KML as the survey name.

Setting a Static IP Address

If connecting over an ethernet connection, a static IP address will need to be set:

1. Open *Network and Internet Settings*.
2. Open *Change Adapter Options*.
3. Open *Properties* for the Ethernet port/adapter.
4. Open *Properties* for IPv4.
5. Set IP address to `10.0.65.2`.
6. Set subnet mask to `255.255.255.0`.
7. Press **OK**.

IP Address Reference

Service	Address
GRYFN WebUI	`10.0.65.50`
HSInsight	`10.0.65.50:8080`
Ouster	`10.0.65.128`
SBG	`10.0.65.100`